

POWER APPS DESIGN: TIPS & TRICKS

Learning responsive design using relative positioning
(Bonus Included! Maximizing the HTML control)



EXPECTATIONS

- **LOVE** questions, but try and stay on topic
- Except during a question, please **keep microphones** muted.
- Okay to interrupt **anytime** you are getting lost or do not understand by raising hand, typing in chat, or just coming off mute
- Some questions might be better suited for parking lot
- If I do not know the answer, I **will find out** and get back to you.
- Questions over microphone **must** be sung as if part of a musical.

HAVE FUN!



WHY SHOULD I CARE?



AGENDA

- 1) Definitions of Terms
- 2) Goals of Presentation
- 3) Goal #1
- 4) Build & Demo
- 5) Goal #2
- 6) Build & Demo
- 7) Wrap-up
- 8) Questions



DEFINITIONS OF TERMS

- **Responsive Layout** – A user interface that automatically adapts to different screen sizes, resolutions, orientations, and content changes to provide a consistent user experience.
- **Relative Positioning** – A design technique where controls are positioned and sized based on other controls or their parent container rather than fixed coordinates, allowing layouts to adjust dynamically.
- **Controls** – Individual user interface elements within Power Canvas Apps.
- **HTML** – HyperText Markup Language.
- **CSS** – Cascading Style Sheets.



GOALS

RESPONSIVE DESIGN

Goal #1

- **Responsive Layout**
- **Relative Positioning:** One technique used to achieve Responsive Layout

HTML LAYOUTS

Goal #2

- **Understnading the Power of HTML Control**
- **String Interpolation**
- **Beautify Design**

RELATIONSHIP

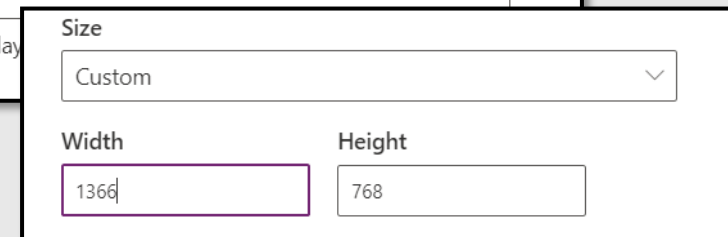
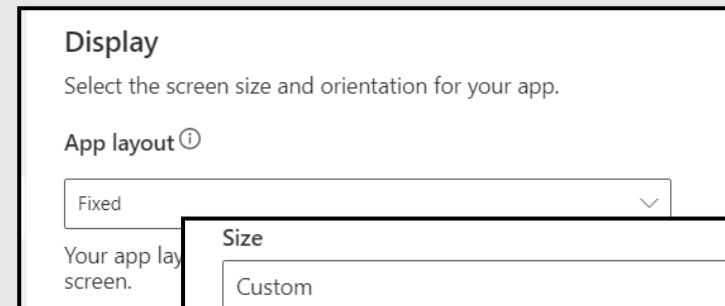
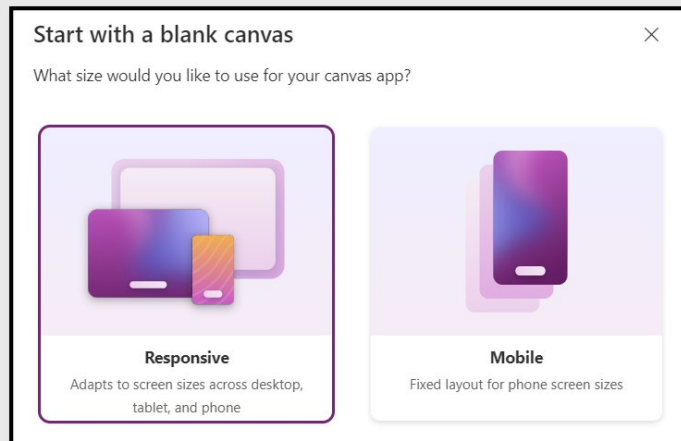


GOAL #1 RESPONSIVE DESIGN

We use Math everyday in development. When working with responsiveness, especially through relative positioning and sizing, Math is a way of life.

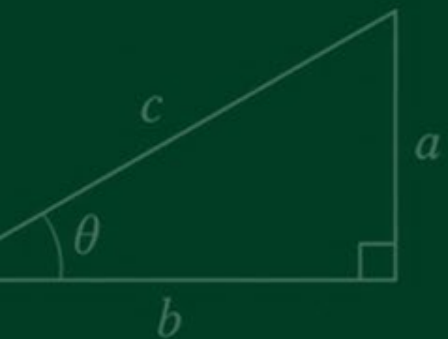
Common Examples

- $(\text{Parent.Width} - \text{Self.Width}) / 2$ **OR** $(\text{Parent.Height} - \text{Self.Height}) / 2$
- $\text{TextInput1.X} + \text{TextInput1.Width} + 15$
- $\text{If}(\text{varShowMe}, \text{TextInput1.X} + \text{TextInput1.Width} + 15, 0)$
- $\text{Image1_1.X} - (\text{Image1.X} + \text{Image1.Width})$
- $\text{Parent.Height} * 0.33$ **OR** $\text{Parent.Height} * 33\%$



$$x^2 + 1 = 0$$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$



$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

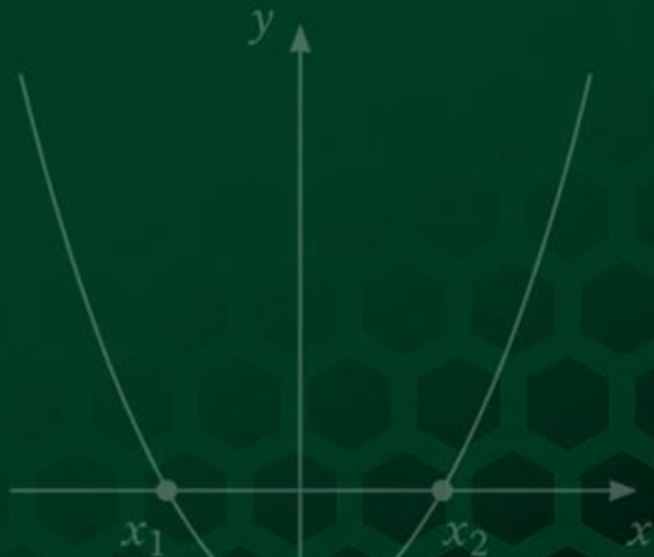
$$ax^2 + bx + c = 0$$

$$b^2 - 4ac \begin{cases} \Delta > 0 & \text{two real roots} \\ \Delta = 0 & \text{one real root} \\ \Delta < 0 & \text{two complex roots} \end{cases}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DEMO TIME!

$$a^2 + b^2 = c^2$$



$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$V = \frac{4}{3} \pi r^3$$



GOAL #2

USING HTML CONTROL



LIGER-TECH Filament Inventory

Total count: 37

	CR-PETG	Condition	Remaining	In-stock
	Creality	Used	330	1
	CR-PETG	Condition	Remaining	In-stock
	Creality	New	330	1
	CR-PETG	Condition	Remaining	In-stock
	Creality	New	330	1
		Used	35.25	1
	CR-PLA	Condition	Remaining	In-stock
	Creality	Used	168.21	1
	CR-TPU	Condition	Remaining	In-stock
	Creality	Used	328	1
	CR-TPU	Condition	Remaining	In-stock
	Creality	Used	328	1

CR-PETG

(Creality)
Date Added: 08-05-2025

Suggested Settings

Nozzle Temp	230°C - 250°C
Bed Temp	70°C - 85°C
Print Speed	30 - 60 mm/s
Adhesion	PEI + glue stick
Enclosure	Not required

Color: Black

Condition	Remaining	On-hand
Used	≈330 m	

Any filament marked as **used** with a date on or before August 3, 2025, reflects approximate usage. Accurate tracking of filament usage begins August 4, 2025.

```
    Grey", ColorsOfFilament.GreyText,
    "Gold", ColorsOfFilament.GoldText,
    "Skin", ColorsOfFilament.SkinText,
    "Purple CF", ColorsOfFilament.PurpleCFText,
    "Glow-in-the-Dark", ColorsOfFilament.GlowInTheDarkText,
    "Metallic Silver", ColorsOfFilament.MetallicSilverText,
    "LightBlue", ColorsOfFilament.LightBlueText,
    "Brown", ColorsOfFilament.BrownText,
    "Red", ColorsOfFilament.RedText
  }' />
</div>
<div>
  <span style='font-size:2em; font-weight:bold;line-height:1.2em'>{galFilamentData.Selected.TypeText}</span><br />
  <span style='font-size:1.25em; line-height:1em;'>{{galFilamentData.Selected.BrandText}}</span><br />
  <em>Date Added: <b>{Text(First(galFilamentData.Selected.GroupedItems).Created,"mm-dd-yyyy")}</b></em>
</div>
<hr />
<div>
  <b>Suggested Settings</b><br />
  <table style='border:1px solid #666;margin-top:.5em; width:{htmlDetails.Width-htmlDetails.PaddingLeft*3-imgLogo.Width}px;
  border-collapse:collapse'>
    <tr style='background:white'><td style='border:1px solid #666; padding:2px 5px;'>Nozzle Temp</td><td style='border:1px
    solid #666; padding:2px 5px;font-weight:600;'>{Lookup(FilamentSettings,Name=galFilamentData.Selected.TypeText,
    NozzleTemp)}</td></tr>
    <tr><td style='border:1px solid #666; padding:2px 5px;'>Bed Temp</td><td style='border:1px
    solid #666; padding:2px 5px;'>{Lookup(FilamentSettings,Name=galFilamentData.Selected.TypeText, BedTemp)}</td></tr>
  </table>
</div>
```

Hyper PLA

(Creality)
Date Added: 08-03-2025

Suggested Settings

Nozzle Temp	200°C - 230°C
Bed Temp	60°C - 75°C
Print Speed	60 - 300 mm/s
Adhesion	PEI/glue stick
Enclosure	Not required





Color: Red

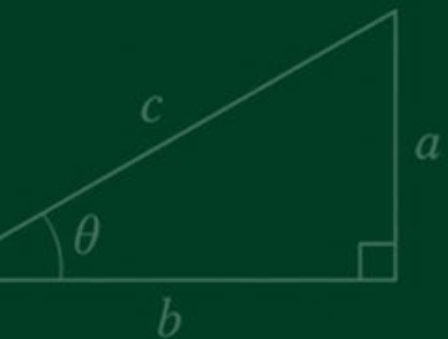
Condition	Remaining	On-hand
Used	≈268 m	1
Used	≈33.5 m	1

Any filament marked as **used** with a date on or before August 3, 2025, reflects approximate usage. Accurate tracking of filament usage begins August 4, 2025.



$$x^2 + 1 = 0$$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$



$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

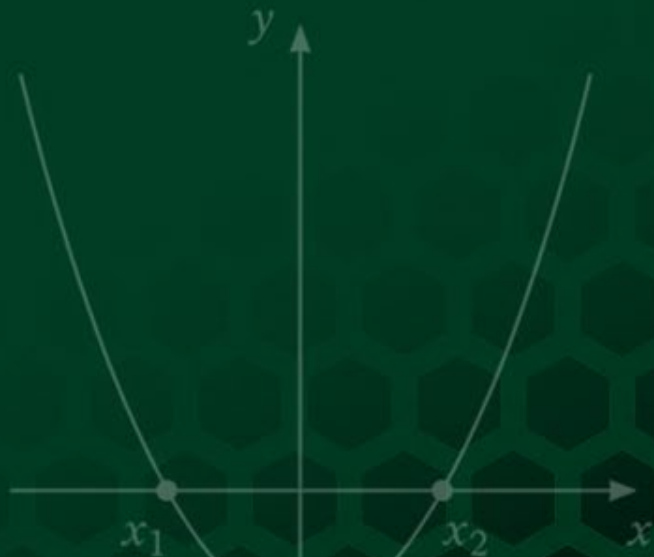
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DEMO TIME!

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$$V = \frac{4}{3} \pi r^3$$



RECAP

GOAL #1

RESPONSIVE DESIGN

- Using Math to set relative position and “motion”
- Scaling image controls

GOAL #2

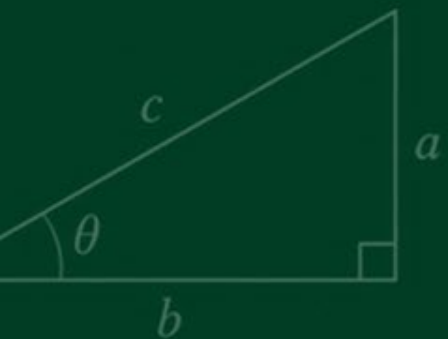
USING THE HTML CONTROL

- Great option for read-only information UI
- Use string interpolation to display dynamic data
- Most modern CSS works!



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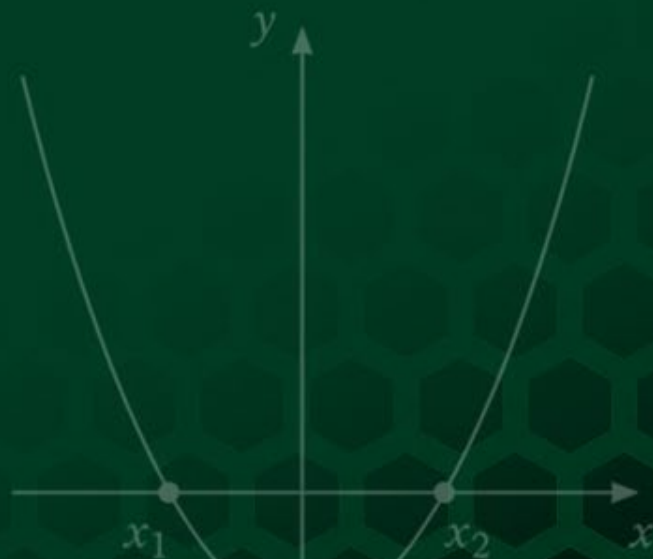
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QUESTIONS?

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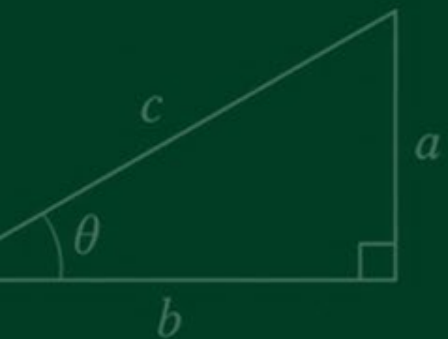
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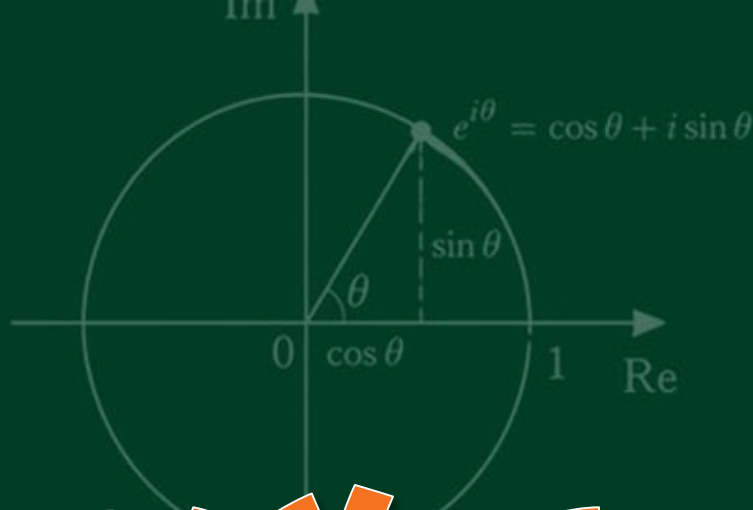
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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a^2 + b^2 = c^2$$

THANK YOU



$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

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